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MILITARY SPECIFICATION

DECK COVERING, LIGHTWEIGHT, NONSLIP, SILICON CARBIDE

PARTICLE COATED FABRIC

1. SCOPE

1.1 This specification covers lightweight, nonslip silicon carbide particle coated fabric deck-covering for weather decks and other locations aboard Naval vessels.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
P-S-606 - Soap-Powder.
U-M-186 - Medicinal Products and Clinical Laboratory Reagents; General Specification for Containers (Packaging and Packing).
TT-E-485 - Enamel, Semi-Gloss, Rust-Inhibiting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-760 - Drums and Pails, Metal (5 and 16.64 Gallon).
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-B-10377 - Boxes: Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-15016 - Lubricating Oil, General Purpose.
MIL-D-16851 - Deck Covering, Lightweight, Nonslip, Nonsparking: and Sealing Compound, Fabric Deck-Covering.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

FSC 7220

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads (40" x 48" 4-Way Partial and 4-Way Pallets).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material. -

3.1.1 The deck covering shall consist of a fabric backing uniformly coated on one surface with silicon carbide particles, approximately number 60 grit (medium grade flint abrasive designations), bonded to the fabric with a suitable resin or similar material. The back of the fabric shall be coated with a pressure-sensitive type of adhesive, covered with a removable protective liner.

3.1.2 Adhesive and liner. - The adhesive shall be pressure-sensitive, coated to cover the back side of the deck covering, and shall adhere immediately to smooth, clean, dry deck surfaces, without wrinkling, curling, breaking, or lifting. A suitable liner shall be used to cover the adhesive film for protection against foreign matter until applied.

3.2 Application. - Deck-covering shall be applicable to deck surfaces such as steel, aluminum alloy, galvanized plates bronze, or wood, shall be watertight, and shall withstand the deteriorating action of oils, gasoline, and alcohol, when such liquids are in contact with the deck for short intervals of time. When applied to a deck surface in an area that is frequently or continuously exposed to direct contact with oil, grease, or gasoline, all edges of the deck-covering shall be protected by a bead of sealer designed to resist any penetration under the deck-covering. Deck-covering shall be capable of being easily and readily applied or laid by the personnel of the Naval shipyards, and easily and readily repaired by the ships' forces.

3.3 Adherence. - Deck-covering, when applied or laid, shall maintain a firm adherence to the deck or structure, and shall protect the metal deck surfaces from corrosion and wooden surfaces from penetration of moisture. It shall form a durable, sanitary, and neat appearing surface.

3.4 Odor. - The deck-covering shall be free from objectionable odors under ordinary service conditions.

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3.5 Liner removal. - When deck-covering is tested as specified in 4.4.2 and 4.4.2.1, the average amount of effort required to pull liner away from the adhesive film shall be as shown in table I.

Table I - Removal of liner.

Condition	Maximum effort required to pull liner away from adhesive film
	Ounces
As received	60
After 7 days aging at 150°F.	60

3.5.1 Liner and adhesive transfer. - When deck-covering is tested as specified in 4.4.2.2, there shall be no transfer of fragments of liner material to the adhesive surface, or adhesive to liner surface, when the protective liner is stripped away at a rate of 1 foot per second.

3.6 Adhesion. -

3.6.1 Shear strength. - The deck-covering, when applied to stainless steel plates, shall conform to table II, when tested as specified in 4.4.3.1.

Table II - Adhesion.

Temperature	Minimum amount of weight in pounds capable of being supported for 1 minute	
	No aging	After 7 days aging
Degrees F.	Pounds	Pounds
160	5	5
80	10	10
32	15	15

3.6.2 Strip strength. - The deck-covering, when applied to high tensile steel and galvanized steel plates, shall conform to table III when tested as specified in 4.4.3.2.

Table III - Strip adhesive strength.

Adhesive strength (load at separation, pounds) (min.)				
Drying time	After 48 hours immersion in water	30°F.	80°F.	160°F.
48 hours	0.75	0.75	0.75	0.75

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3.7 Thickness and weight. - Deck-covering shall be not greater than 0.080 inch thick when applied to the steel decks, and shall weigh not more than 5 ounces per square foot (see 4.4.4).

3.8 Resistance to elevated temperature. - Deck-covering shall not flow nor slip in any part more than 1/16 inch, nor soften, when tested as specified in 4.4.5.

3.9 Resistance to detergent solution. - Deck-covering shall not soften, stain, streak, nor show any appreciable loss in color when cleaned at regular intervals with detergents such as soap-powder conforming to Specification P-S-606 (see 4.4.6).

3.10 Nonslip properties. - When tested as specified in 4.4.7, the material shall show factors of friction not less than those shown in table IV.

Table IV - Nonslip properties.

Property	Contacting surface condition		
	Dry	Wet	Oily
Factor of static friction:			
Leather	0.60	0.60	----
Rubber	.60	.60	0.60
Factor of sliding friction:			
Leather	.40	.40	----
Rubber	.50	.60	0.30

3.11 Resistance to accelerated corrosion. - Deck-covering shall not soften or become detached, and there shall be no corrosion on the surface of the covering, or on the surface of the steel beneath the deck-covering, when tested as specified in 4.4.8.

3.12 Resistance to moisture and temperature changes. - When tested as specified in 4.4.9, deck-covering shall show no signs of cracking, separation from the steel plate, or corrosion on the surface of the steel plate beneath the coating.

3.13 Resistance to oil. - The deck-covering applied and beaded with sealer shall show no signs of softening, loss of adhesion, or any other form of deterioration after 24-hour immersion in lubricating oil conforming to symbol 3042 of Specification MIL-L-15016 (see 4.4.10).

3.14 Fire resistance. - The deck-covering, when tested in accordance with 4.4.11, shall exhibit linear damage no greater than 8 inches. The combustion plus ignition time shall not exceed 4 minutes.

3.15 Size. - Deck-covering shall be furnished in treads 6 by 24 inches, 8 by 30 inches, or in rolls 1-1/2, 6, 8, 14, and 24 inches wide and 96 feet long, as specified (see 6.1).

3.16 Wearing life. - The deck-covering shall show wear not to exceed 0.030 inch after 1,500 revolutions, when tested as specified in 4.4.12.

3.17 Color. - When color is required for deck-covering, the color shall be as specified (see 6.1).

3.18 Workmanship. - Workmanship shall be first class in every respect.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling. -

4.2.1 Lot. - For purpose of sampling, a lot shall consist of not more than 10,000 square feet of deck-covering offered for inspection at one time.

4.2.2 Sampling for visual and dimensional examination of treads. - From each lot of deck-covering the inspector shall select at random sample of treads with lot acceptance based on table V in accordance with Standard MIL-STD-105.

Table V - Sampling for lot inspection of treads
AQL (approx.) = 2.5 percent defective.

Number of deck-covering treads in lot	Number of deck-covering treads in sample	Acceptance number (defectives)	Rejection number (defectives)
110 and under	10	0	1
111 to 300	15	1	2
301 to 500	25	1	2
501 to 1300	35	2	3
1301 to 3200	50	3	4
3201 and over	75	4	5

4.2.3 Sampling for visual and dimensional examination of material in rolls. - From each lot of deck-covering the inspector shall select at random sample of treads or rolls with lot acceptance based on table VI in accordance with Standard MIL-STD-105.

Table VI - Sampling for lot inspection of rolls
AQL (approx.) = 2.5 percent defective.

Number of rolls in lot	Number of rolls in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	1	2
110 and over	25	1	2

4.2.4 Sampling for tests. - From each lot shall be selected a representative 6 x 24 inch sample for tests. The samples selected shall be subjected to the tests specified in 4.4. If any sample fails any test the lot shall be rejected. Rejected lots may be offered again for Government tests provided the contractor has reworked the entire lot so as to remove or correct all nonconforming material.

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4.3 Visual and dimensional examination. - Each of the sample treads or rolls selected in accordance with 4.2.2 and 4.2.3 shall be visually and dimensionally examined to determine conformance with the requirements of this specification which do not require tests. Any roll or tread in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective rolls or treads in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for examination provided that the contractor has repaired or removed all nonconforming rolls or treads. The inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

4.4 Test procedures. -

4.4.1 General. - The surface of the steel plates to which the deck-covering is attached shall be free from all rust, scale, and organic matter. The steel samples with deck-covering applied, including beading sealer shall be cured for 48 hours at room temperature before conducting tests. Unless otherwise specified in the contract or order, all tests are referred to atmospheric conditions at a temperature of 70° to 75° Fahrenheit (F.) and relative humidity of 65 ± 2 percent.

4.4.2 Liner removal. - Twenty specimens 1 by 7 inches shall be prepared from each lot of deck-covering. Ten specimens shall be tested immediately and ten specimens placed in an oven at 150° ± 5°F. for 7 days, cooled for 2 hours, and then tested in accordance with 4.4.2.1. The amount of effort required to pull liner away from adhesive film shall be determined for each of the ten specimens, the results averaged, and the average of the ten results shall not exceed the maximum values specified in table I.

4.4.2.1 Apparatus and procedure. - The apparatus shall be a pendulum type tensile testing machine. The test shall be made with the stressing jaw moving at a speed of 12 inches per minute. The liner of the 1- by 7-inch specimen of deck-covering shall be partially removed from one end sufficiently so that the free liner end may be placed in the stressing jaw and the deck-covering end placed in the fixed jaw. The maximum tension required to remove the remainder of the liner shall be taken as the amount of removal effort required for each specimen. The pawls of the pendulum arm shall be taped back in an open position. The scale shall be graduated to record results in ounces.

4.4.2.2 Liner and adhesive transfer. - The protective liner of a 6- by 24-inch tread shall be pulled away from the deck-covering at a speed of 1 foot per second. The liner and the decking material shall then be examined for signs of transfer of adhesive to the liner surface and for retention of liner fragments by the adhesive surface.

4.4.3 Adhesion. -

4.4.3.1 Shear strength. - Thirty specimens 1 by 7 inches shall be prepared from each lot of deck-covering. Fifteen specimens shall be tested immediately and 15 specimens conditioned in an oven at 150° ± 5°F. for 7 days, and tested in accordance with 4.4.3.1.1.

4.4.3.1.1 Procedure. - The protective liner shall be stripped back approximately 2 inches from one end of each 1- by 7-inch specimen as it is to be tested, care being taken to avoid any foreign matter coming in contact with the exposed adhesive film. With the use of a template such as shown on figure 1, the exposed end shall be adhered to a polished clean stainless steel plate so that exactly 1 square inch of the specimen has contact. The plate is then inserted in a jig and pressure applied by any suitable hydraulic or mechanical press to provide 65 pounds per square inch of pressure, in mounting test specimens. The specimen shall remain under this pressure for 1 minute. The 15 specimens aged in oven at 150°F. for 7 days, and the 15 specimens without aging shall be mounted in the above manner, and allowed to set at room temperature for 48 hours after which time five each of aged and unaged specimens shall be tested for adhesion at room temperature (80° ± 5°F.). Five each of aged and unaged specimens shall be tested for adhesion at 160° ± 5°F. and five each of aged and unaged specimens shall be tested at 32° ± 2°F. The elevated temperature 160°F. shall be deemed to have been reached after conditioning the specimen in an oven for 1 hour. The lower temperature of 32°F. shall be deemed to have been reached after conditioning in a cold chamber for 1 hour. The tests shall be made by fixing the steel plate in a vertical position leaving 6 inches of the specimen strip below to which a weight can be attached and suspended as shown on

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figure 2. Weight of the amount specified in table II shall be placed in the weight pan. The deck-covering shall be considered to have passed this test if at least three of each group of five specimens have been capable of supporting the weights specified for a period of 1 minute.

4.4.3.2 Strip strength. - Eight samples of high tensile steel plates and 8 samples of galvanized steel plates each 10 by 16 inches by 1/8 inch thick, shall be prepared by buffing the surface on one side with a power driven flexible backed number 24 abrasive disk using such pressure as will just polish the surface of the metal. The buffed plates shall then be cleaned with a solvent. The deck-covering with protective liner removed, shall be applied to the clean steel plates by rolling fives times in each direction with a roller traveling at approximately 7-1/2 feet per minute. The roller, as shown on figure 3 and figure 4 is composed of steel disks 7 inches in diameter and approximately 1/4 inch in thickness. The disks have a central hole 2 inches in diameter and are assembled on a 1/2-inch diameter rod, with a disk having a 5/8 inch hole fastened on each end. Accordingly, when the roller is pulled along by the handle the disks are free to aline themselves vertically such that they conform to local irregularities of the specimen surface in 1/4 inch steps. The 7-inch diameter roller results in a loading of 10 pounds per inch of width which approximates 65 pounds per square inch.

4.4.3.2.1 Procedure. -

4.4.3.2.1.1 Adhesion at room temperature (80° ± 5°F.). - Three specimens attached to high tensile steel and three specimens attached to galvanized steel plate prepared as specified in 4.4.3.2 shall be employed. The specimens shall be allowed to dry for a total of 48 hours, and immediately tested. The method of testing strip adhesion to metal shall be as shown on figure 5. Adhesive strength shall be taken as the highest load at which the separation in 1 minute does not exceed 1 inch. The average of the two highest values obtained on the three strips shall then be considered as the adhesive strength between the deck-covering and the metal.

4.4.3.2.1.2 Adhesion after 48 hours immersion in water. - Duplicates of specimens prepared as specified in 4.4.3.2 shall be immersed in water for 48 hours and immediately tested in the same manner.

4.4.3.2.1.3 Adhesion at elevated temperature (160° ± 5°F.). - Duplicates of specimens prepared as specified in 4.4.3.2 shall be immediately tested with the metal back of the test specimen in contact with the 6- by 12-inch electric heating unit controlled by a rheostat to maintain even temperature during test. Temperature shall be determined by means of a thermocouple inserted between the cloth and the metal plate in the center of the strip. The cloth shall be raised from the metal by inserting a 1/16 inch awl a distance of 1 inch at an angle of 40 degrees parallel to the 12-inch side of the panel; 1/8 inch hole punched in the center of the cloth shall act as a starting base for placement of the awl. The point of the awl shall be moved along the surface of the metal the required 1 inch distance. The thermocouple shall be slipped into position between the raised cloth and the metal panel under the awl and the awl removed. The temperature of the panel at the metal to cloth contact shall be raised to 160°F. and maintained at this temperature for 15 minutes as shown on figure 6, before proceeding with the adhesion test as shown on figure 5.

4.4.3.2.1.4 Adhesion at low temperature (30° ± 2°F.). - Duplicates of specimens prepared as specified in 4.4.3.2 shall be mounted as shown on figure 6, except that the metal back of the test panel shall contact a 6- by 12-inch pan which has been filled with any ice and salt mixture in order to obtain a temperature of 30°F. at the metal to cloth contact. This temperature shall be maintained for 15 minutes as shown on figure 6 before proceeding with the adhesion test as shown on figure 5.

4.4.4 Thickness and weight. - The deck-covering shall be applied to three 6-inch square, 0.025 inch (25 mils) thick steel plates which have been previously measured and weighed. When the material has set for 48 hours, the 3 test specimens including the steel plates, shall each be weighed to the nearest 0.1 ounce. The length and width shall be measured to the nearest 0.01 inch and the thickness to the nearest 0.001 inch. From the difference between the weight of the coated steel plate and the uncoated steel plate, the weight of the material shall be computed in ounces per square foot per thickness as applied. The thickness shall be determined by measuring the steel plates, with and without the coating, at 16 equally distributed points on the specimen, by means of a dial thickness gage and a template. The difference in thickness of the mounted steel plate and the coated steel plate shall be averaged to determine the thickness of the material. The final weight shall be the average of the 3 specimens.

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4.4.5 Resistance to elevated temperature. - The resistance of the deck-covering to elevated temperature shall be determined as follows:

4.4.5.1 Flow and slip. - Specimens of the deck-covering 6 by 2 inches shall be scribed with a line parallel to and approximately 1 inch from a 2-inch edge used as a reference. The distance between this line and the edge of the steel plate shall be measured to the nearest 0.01 inch. The specimen shall then be suspended vertically from the end opposite the reference end in an oven maintained at a constant temperature of 158°F. for 5 hours. When the sample has cooled to room temperature, the distance between the reference edge and the line shall be measured again. The difference between the 2 measurements is called the flow and slip.

4.4.5.2 Softening. - The deck-covering shall be examined by touch, immediately after the sample has been removed from the oven, to determine whether the material has softened under the action of heat.

4.4.6 Resistance to detergent solution. - Specimens of deck-covering 6 by 2 inches shall be immersed for 1 hour in the solution specified below so that a 3-inch section of the specimen is immersed in the solution and the upper half remains unexposed for purposes of comparison. The liquid solution shall be made by dissolving soap powder, conforming to Specification P-S-606, in tepid water to form a 1/2 of 1 percent solution, then adding lamp-black until the solution is dark gray. After immersion, the specimen shall be rinsed for 1 minute with tap water at 140°F. and dried with an air jet. The specimen shall then be examined for softening, stains, streaking, or loss of color.

4.4.7 Nonslip properties. - The factors of friction of the deck-covering against leather and rubber shall be determined, respectively. The leather shall be oak-tanned sole leather which has been sanded smooth with grade 0 garnet paper. The rubber shall be a vulcanized compound with a hardness range of 60 to 80 Durometer "A". Tests shall be made with the contact surfaces dry; wet with a solution of 4 percent sodium chloride in water; and oiled, using oil conforming to symbol 3042 of Specification MIL-L-15016. The leather and rubber shall be cut to a size 2 by 4 inches and mounted on a block of the same size. A load of 33 pounds shall be applied uniformly over the 2- by 4-inch block during the test. A pendulum-type tensile testing machine shall be used to measure the force required to move the weight block on the deck-covered surface. The moving jaw of the tensile testing machine shall be set to move at the rate of 7-1/2 feet per minute. Approximately one cup of the salt water solution shall be poured over the deck-covering surface in making the wet test. For the oil test the oil shall be applied to the center of the leather or rubber sample and on the deck-covering surface sufficient to spread evenly over the entire surface.

4.4.8 Accelerated corrosion test. - Steel plates, 6 by 2 inches covered with the decking material, shall be immersed in a 10 percent salt ($NaC\ell$) solution for 15 days, during which time a continuous stream of air shall be passed through the solution in order to promote corrosion. All exposed metal parts of the steel plates shall be coated with zinc chromate primer.

4.4.9 Resistance to moisture and temperature changes. - Tests shall be made on specimens 6 by 2 inches. All exposed parts of the steel shall be coated with zinc chromate primer. The specimens shall be immersed in a solution of 4 percent sodium chloride in water, under a pressure of 8 pounds per square inch, for 48 hours. Immediately after immersion, the specimens shall be subjected to 2 complete cycles of alternate exposure to a temperature of 0° ± 5°F., for 24 hours, followed by a temperature of 120° ± 5°F., for 24 hours. The specimens shall then be examined for evidence of cracking or other failure. A portion of the deck-covering shall be carefully removed from the plate to observe any signs of rusting or corrosion under the deck-covering.

4.4.10 Resistance to oil. - Two steel specimens 6 by 6 inches with deck-covering applied and beaded with sealer conforming to Specification MIL-D-16651 shall be immersed in lubricating oil conforming to symbol 3042 of Specification MIL-L-15016 for 24 hours. The deck-covering and beading sealer shall be examined for softening, loss of adhesion or any other deterioration.

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4.4.11 Fire-resistance test. -

4.4.11.1 Apparatus. - The fire-resistance of the deck-covering shall be determined by an apparatus as shown on figure 7, which consists of communicating horizontal and vertical flues constructed of asbestos board sheets supported on an angle-iron frame, with the exception of the horizontal bottom plate, which is steel. The enclosure shall be 36 inches long overall; the flue shall be 8 inches wide by 6 inches high. The bottom plate shall be cut away 3 inches from the firing end to provide clearance for the flames of four open blast gas burners. Details of the fire-resistance apparatus are shown on figures 8 and 9.

4.4.11.2 Procedure. - The deck-covering shall be prepared in the usual manner on a 1/8 inch mild steel plate 31-1/2 inches long by 7 inches wide, with longitudinal flanges for mounting in the horizontal flue. The plate shall be secured in the flue, 2 inches above the bottom plate and with the inner end spaced 3 inches from the flue end, to permit hot gases passing beneath the test plate to be vented through the vertical flue. The four open blast burners shall be located side by side and parallel to the front end of the test plate, on 1-3/4-inch centers, equidistant from each side of the flue. The center of the burners shall be located 4 inches below the bottom surface of the test plate, and shall be 1/2 inch in front of the firing end of the test plate. The gas used shall be commercial propane having a heating value of 2550 British thermal unit (B.t.u.) per cubic feet at a temperature of 60°F. and atmospheric pressure of 30 inches of mercury. The gas and air shall be supplied to a common manifold at a rate of 9.6 and 150 cubic feet per hour respectively, when referred to standard conditions of 60°F. and 30 inches of mercury. The burner flames shall be applied to the test plate for four minutes and then immediately removed. A draft of 0.06 (+0.01 -0.00) inch of water shall be maintained in the flue of the apparatus for the duration of the test.

4.4.11.2.1 Combustion plus ignition time, and linear damage shall be noted. These quantities are defined as follows:

(a) Combustion plus ignition time. - The time taken from the initial application of burner flames until all flaming of the specimen ceases.
(b) Linear damage. - The length of specimen permanently damaged by burning, charring, blistering or embrittlement. This measurement shall be taken after all flaming ceases.

4.4.12 Wearing life. - Samples, 3 inches long by 2 inches wide, shall be prepared so that an area of approximately 6 square inches is presented to the abrasive in the machine specified in 4.4.12.1. The abrasive grit shall be number 80 aluminum oxide. Before wear-testing, the thickness shall be measured at 12 equally distributed points on the samples by means of a dial thickness gage and a template. The sample then shall be subjected to 500 revolutions and the thickness obtained at the same points previously measured.

4.4.12.1 Wear-test machine. - The wear-test machine as shown on figures 10 and 11 shall have the following essential mechanical characteristics:

(a) Mild steel, abrasion disk, 14-1/8 inches in diameter and one-half inch thick, revolved clockwise at a constant speed of 23.5 revolutions per minute (r.p.m.).
(b) Specimen holder, 2 by 4 inches attached to a 5/8 inch diameter shaft by means of a universal coupling. The distance between centers of the disk and the specimen holder shaft is 5.1 inches. A specimen of decking material is secured in the specimen holder by means of clamps at each end of the holder. The holder revolves clockwise at a constant speed of 32.5 r.p.m.
(c) Hardened tool-steel cam attached to the specimen holder shaft by means of set-screws. In each one-half revolution of the shaft, the specimen is lifted a distance of one-sixteenth inch and then dropped back on disk (a) under a 10-pound weight. Change in the lift due to wear of the specimen is compensated by adjustment of the cam at regular intervals, so that the lift is maintained at a practically constant distance.
(d) Weight, 10 pounds, the maximum found suitable for use with this test machine.

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(e) Distributor for abrasive grits. The grits flow by gravity from the hopper (f) through a 13/64 inch inside diameter tube and into the center of the distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grits through eight 3/64 inch diameter holes centered on a 1-5/8 inch diameter pitch circle. A uniform distribution of grits falls to disk (a) in the path of the test specimen. The distributor is revolved at a constant speed of 9 r.p.m.
(f) Hopper for feeding abrasive grits.
(g) Bevel drive gear.
(h) Predetermining revolution counter and switch.
(i) One-third horsepower compound wound direct current motor.
(j) Hopper for collecting used abrasive grit.
(k) Steel frame.

Details of the distributor for the abrasive grit, the cam and specimen holder are shown on figures 12 and 13, respectively.

4.4.12.2 Procedure. - The steel disk (a) revolves while the weighted specimen of deck material in holder (b) is lifted and dropped by a cam while it is being revolved. When the specimen is dropped through motion of the cam, it falls a distance of one-sixteenth inch and meets disk (a) with impact simulating the fall of a foot on the deck surface. Foot action is further simulated in that the revolving specimen contacts the disk with a sliding and twisting motion until lifted clear by the cam. Continuous rotation of the specimen introduces the element of wear in all directions on the decking material. Free abrasive grits of number 80 aluminum oxide are fed at a steady rate from the hopper (f) and distributed uniformly on the steel disk in the path of the specimen. The revolving motions of the steel disk and the test specimen keep the abrasive grits circulating to the outer side of the path of the specimen. The grits fall from the edge of the steel disk into hopper (i), from which they are collected and then screened to eliminate undesired particles before replacement in the upper feeding hopper (f).

4.5 Inspection of preparation for delivery. - Sample packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance to the requirements of section 5 herein.

4.6 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - Deck-covering and beading sealer shall be packaged as follows:

5.1.1.1 Deck-covering. - Deck-covering of like kind and size and in quantities specified (see 6.1) shall be packaged in fiber boxes conforming to Specification PPP-B-636 (class 2). Box closure shall be as specified in the appendix to the box specification. All corner and edge seams and manufacturer's joints shall be sealed with a minimum 1-1/2 inch wide tape conforming to Specification PPP-T-76.

5.1.1.2 Sealer. - Sealer shall be furnished in 5-ounce metal collapsible tubes, 1-gallon cans or in 5-gallon pails as specified (see 6.1).

5.1.1.2.1 Collapsible tubes. - Five-ounce collapsible tubes shall conform to type II, class B of Specification U-M-186. Each tube shall be placed in folding or set-up paperboard box conforming to Specification PPP-B-566 or PPP-B-676 at the option of the contractor. Box closure shall be as specified in the appendix to the applicable box specification.

5.1.1.2.2 One-gallon cans. - One-gallon cans shall conform to type V, class 2 of Specification PPP-C-96. Exterior plan B coating and side seam striping shall be required. Cans shall be provided with wire handles which shall be galvanized or protectively coated to resist corrosion. After filling, the plugs shall be spot soldered to the friction ring at three points, spaced equidistant from each other around the periphery of the plug.

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5.1.1.2.3 Five-gallon pails. - Five-gallon pails shall conform to type II of Specification PPP-D-760. Exterior coating of pails shall be required and shall conform to Specification TT-E-485, color olive drab.

5.1.2 Level C. - Deck-covering and sealer shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. -

5.2.1 Level A. -

5.2.1.1 Deck covering. - Deck covering shall be packed in wood-cleated fiberboard, nailed wood, wirebound, fiber, wood-cleated veneer paper overlaid or wood-cleated plywood boxes conforming to Specification PPP-B-591 (overseas type), PPP-B-621 (class 2), PPP-B-585 (class 2 or 3), PPP-B-636 (class 3), MIL-B-10377 (overseas type), or PPP-B-601 (overseas type), respectively at the option of the contractor. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto. In lieu of steel strapping, fiber boxes may be banded with tape conforming to type IV of Specification PPP-T-97 and the appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Unit containers conforming to class 2 of Specification PPP-B-636 closed, sealed and banded as specified herein and 5.1.1.1 and used as the shipping container need not be overpacked.

5.2.1.2 Sealer. -

5.2.1.2.1 Collapsible tubes. - Collapsible tubes shall be packed for export shipment in accordance with the requirements of Specification U-M-186.

5.2.1.2.2 One- and 5-gallon containers. - One-gallon cans shall be arranged and packed for overseas shipment in accordance with the appendix to Specification PPP-C-96. Five-gallon pails shall require no additional packing. When specified (see 6.1), 5-gallon pails may be palletized.

5.2.2 Level B. -

5.2.2.1 Deck covering. - Deck covering shall be packed in wood-cleated fiberboard, nailed wood, wirebound wood, wood-cleated plywood, wood-cleated veneer paper overlaid or fiber boxes conforming to Specification PPP-B-591 (domestic type), PPP-B-621 (class 1), PPP-B-585 (class 1), PPP-B-601 (domestic type), MIL-B-10377 (domestic type) or PPP-B-636 (class 2), respectively at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. Steel strapping of class 2, Specification PPP-B-636 fiber boxes shall not be required. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Unit containers conforming to Specification PPP-B-636 (class 2) closed as specified herein and sealed as specified in 5.1.1.1 and used as the shipping container need not be overpacked.

5.2.2.2 Sealer. -

5.2.2.2.1 Collapsible tubes. - Collapsible tubes shall be packed for domestic shipment in accordance with Specification U-M-186.

5.2.2.2.2 One- and 5-gallon containers. - One-gallon cans shall be arranged and packed for domestic shipment in accordance with the appendix to Specification PPP-C-96. Five-gallon pails shall require no additional packing. When specified (see 6.1), 5-gallon pails may be palletized.

5.2.3 Level C. - Deck covering and sealer shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers or method of shipment shall comply to the rules and regulations as applicable to the mode of transportation.

5.3 Pallets. - When specified (see 6.1), 5-gallon pails shall be palletized in accordance with Standard MIL-STD-147.

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5.4 Marking. - In addition to the labeling specified in 3.22.7 and any special marking required by the contract or order, unit and interior packages, shipping containers and palletized unit loads shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of treads or width or rolls required (see 3.15).
(c) Color of deck-covering, when required (see 3.17).
(d) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(e) Whether 5-gallon pails shall be palletized (see 5.3).
(f) Quantity per unit package (see 5.1.1.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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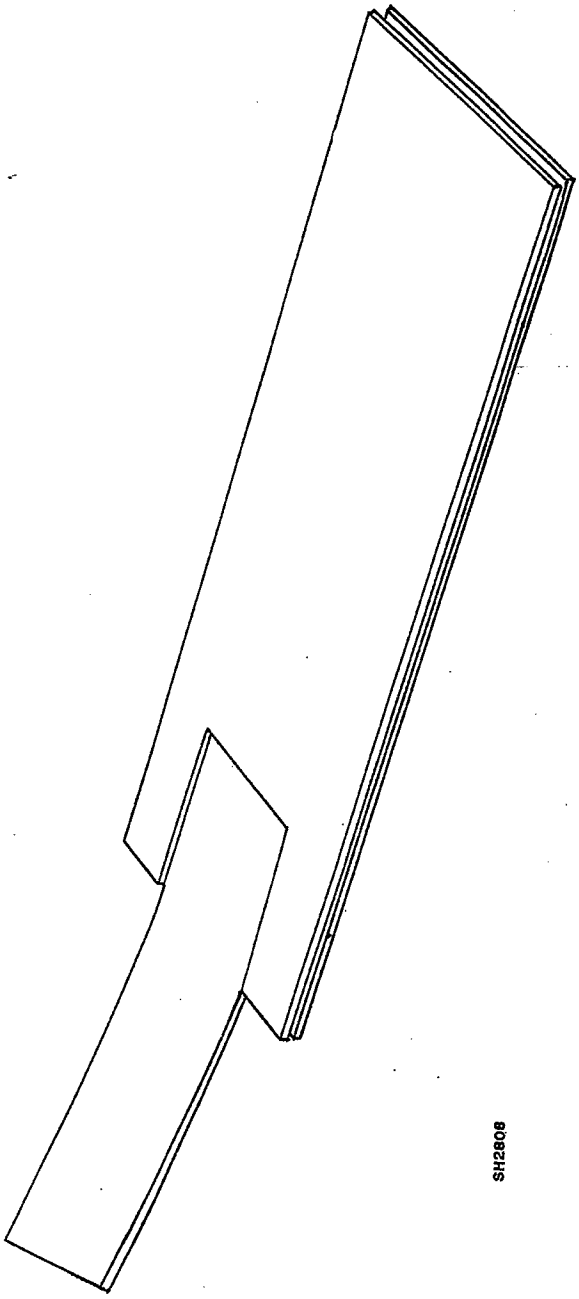


Figure 1 - Template.

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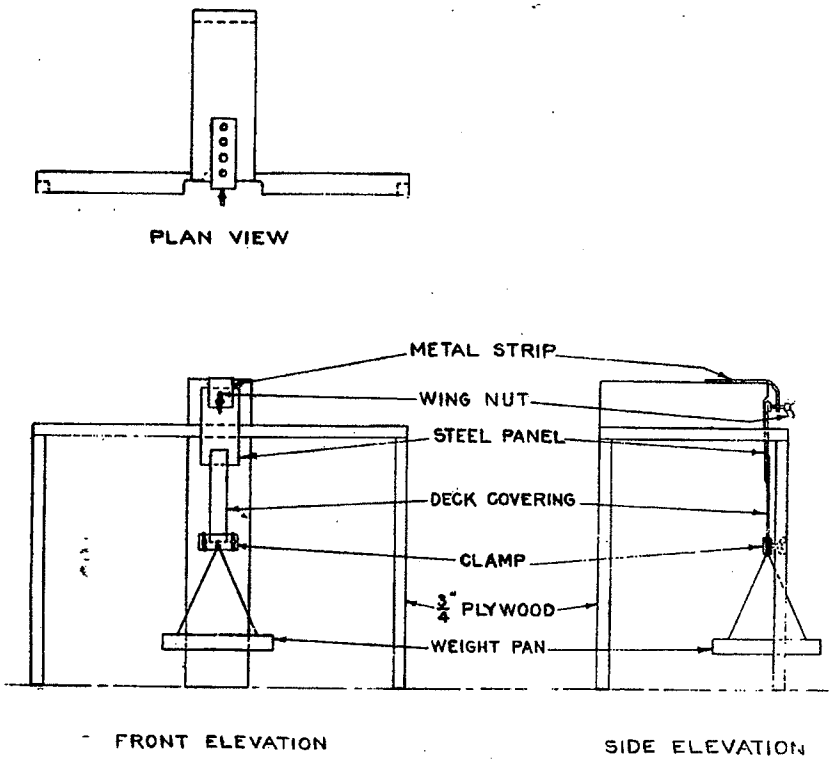


Figure 2 - Adhesion testing machine.

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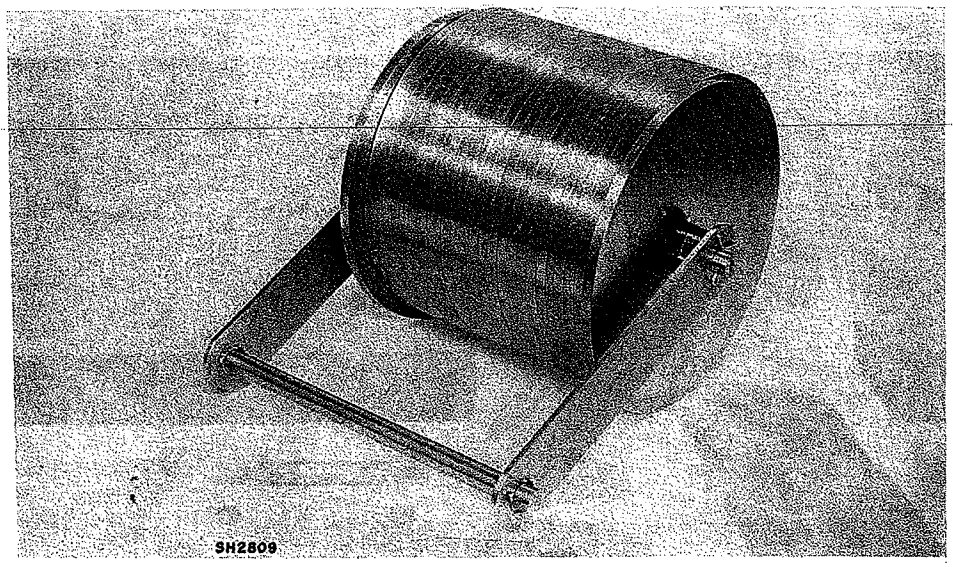
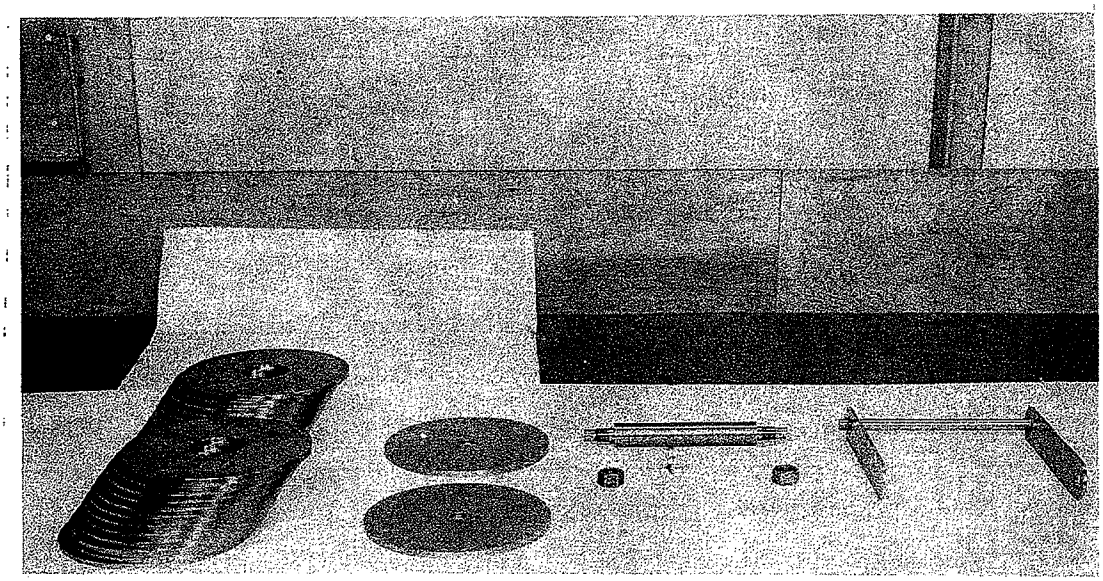


Figure 3 - Multidisk roller for preparation of strip adhesion strength specimens assembled view.

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Figure 4 - Multidisk roller for preparation of strip adhesion strength specimens disassembled view.

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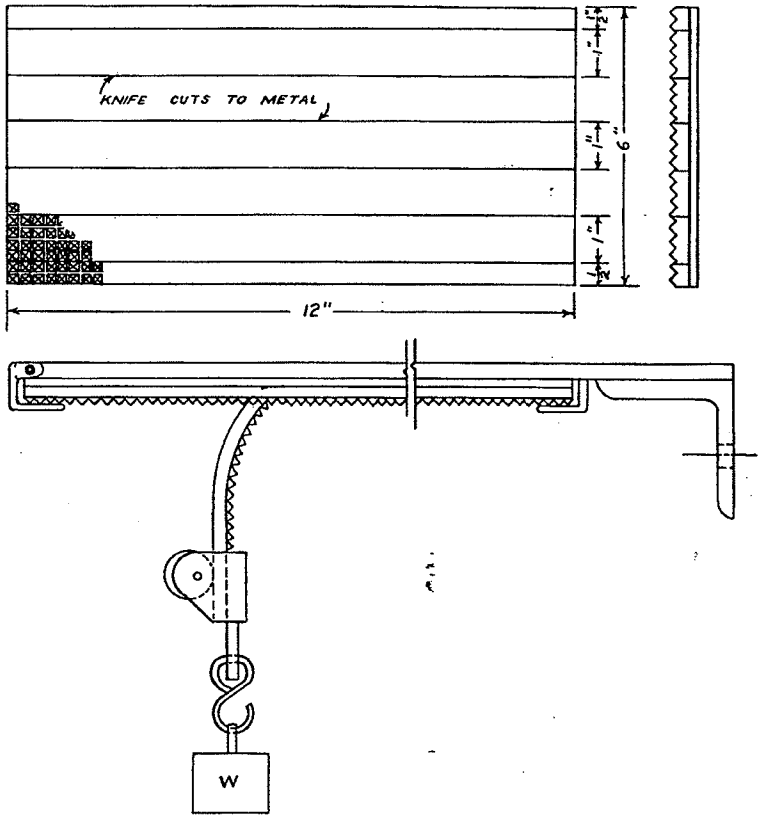


Figure 5 - Strip adhesion test.

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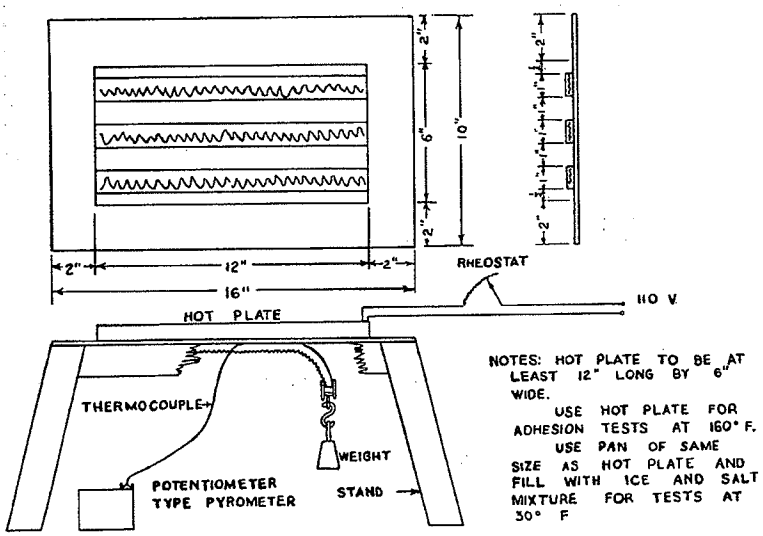
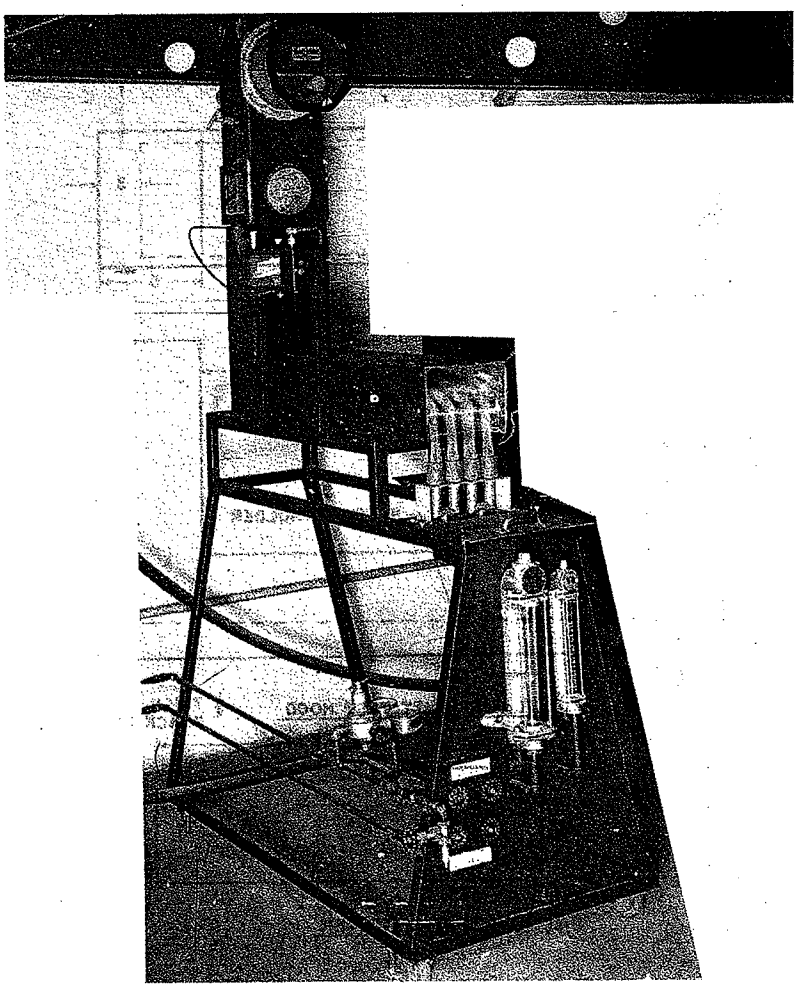


Figure 6 - Apparatus for testing adhesion to steel at elevated and low temperatures.

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Figure 7 - Fire-resistance test apparatus.

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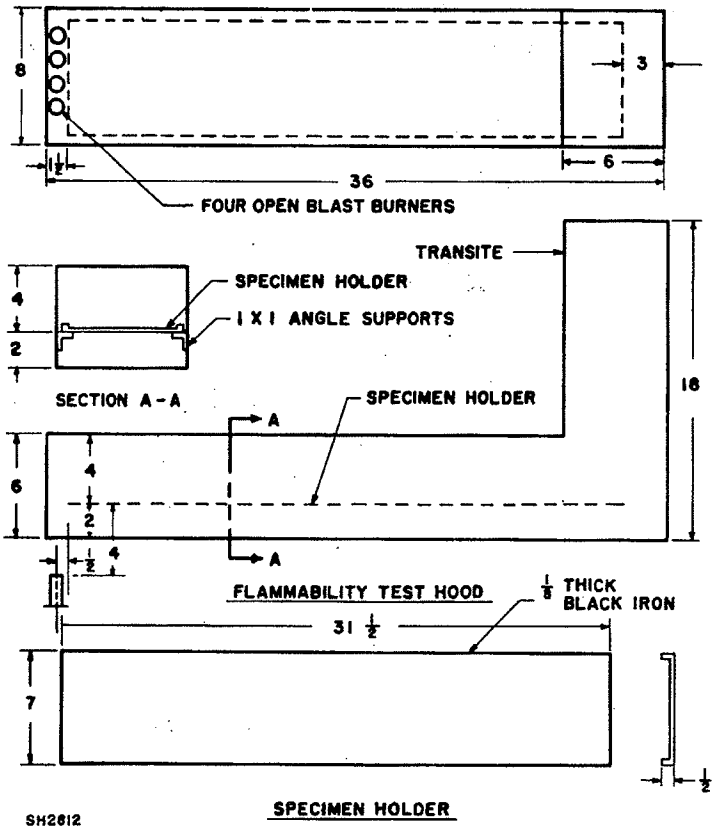


Figure 8 - Detail of fire-resistance test apparatus.

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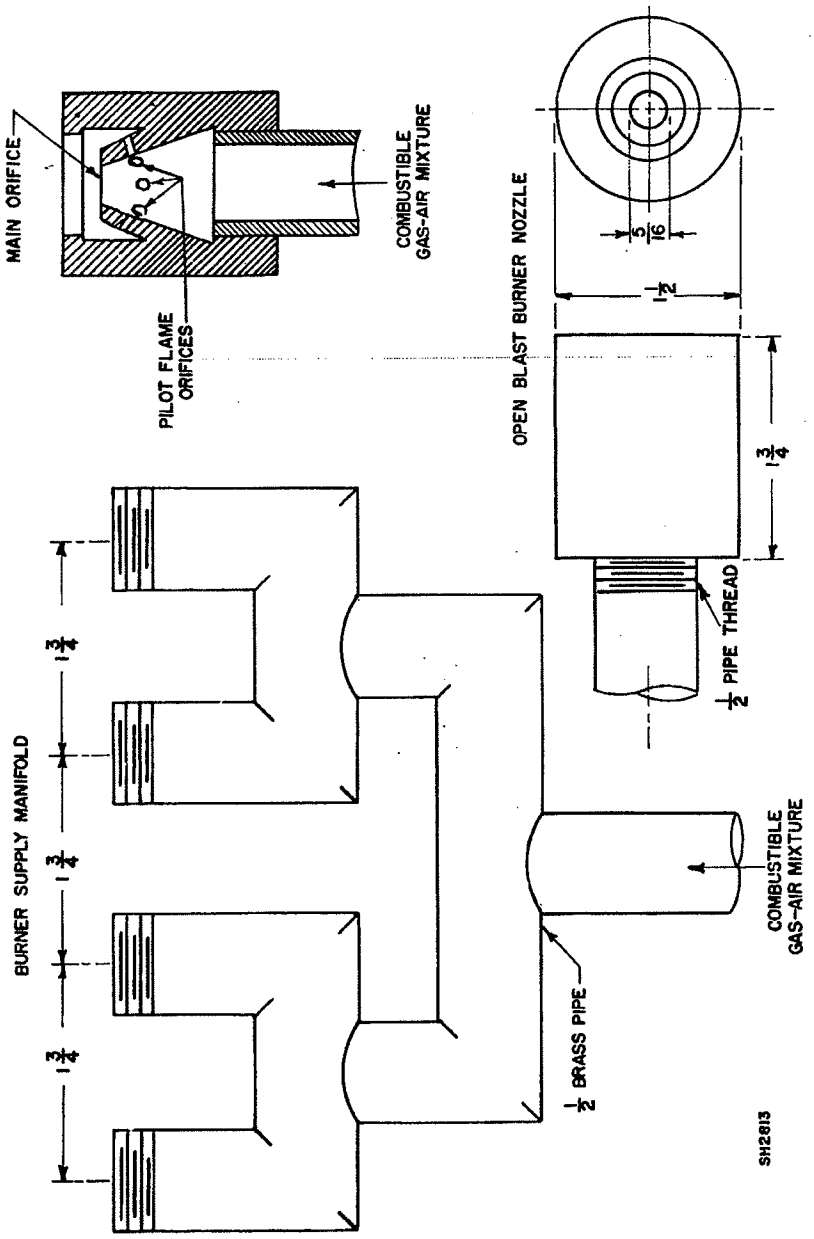


Figure 9 - Burner supply manifold and open blast burner nozzle.

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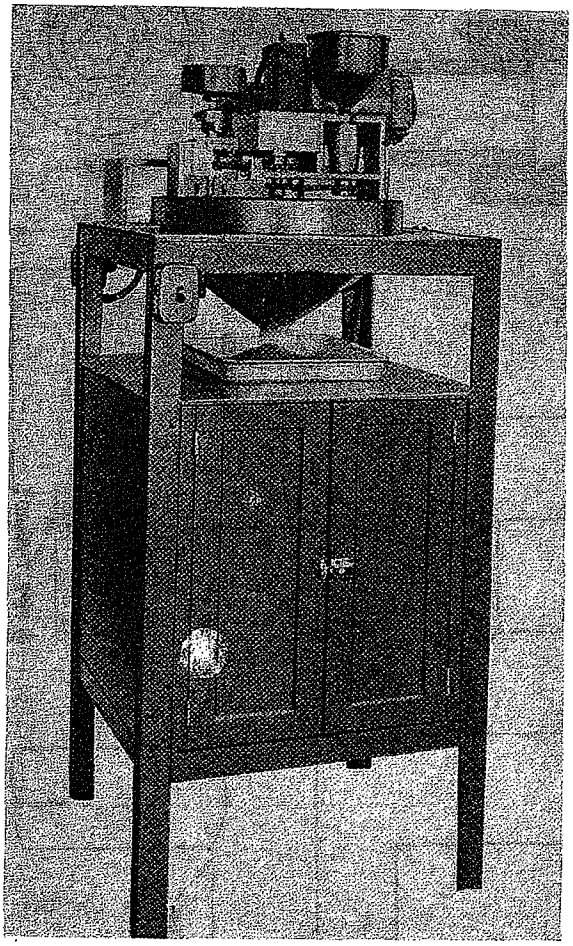


Figure 10 - Wear-test machine.

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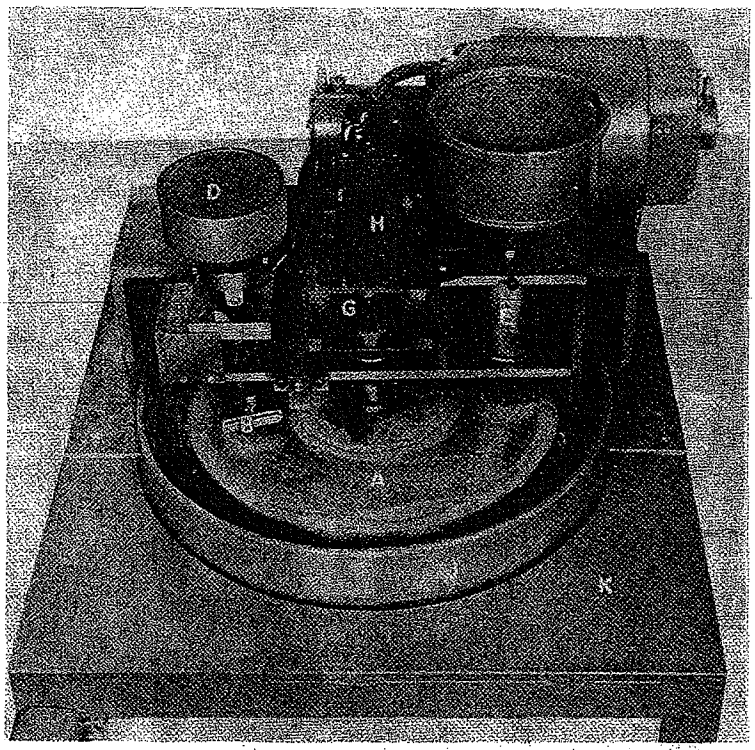


Figure 11 - Detail view of water-test machine.

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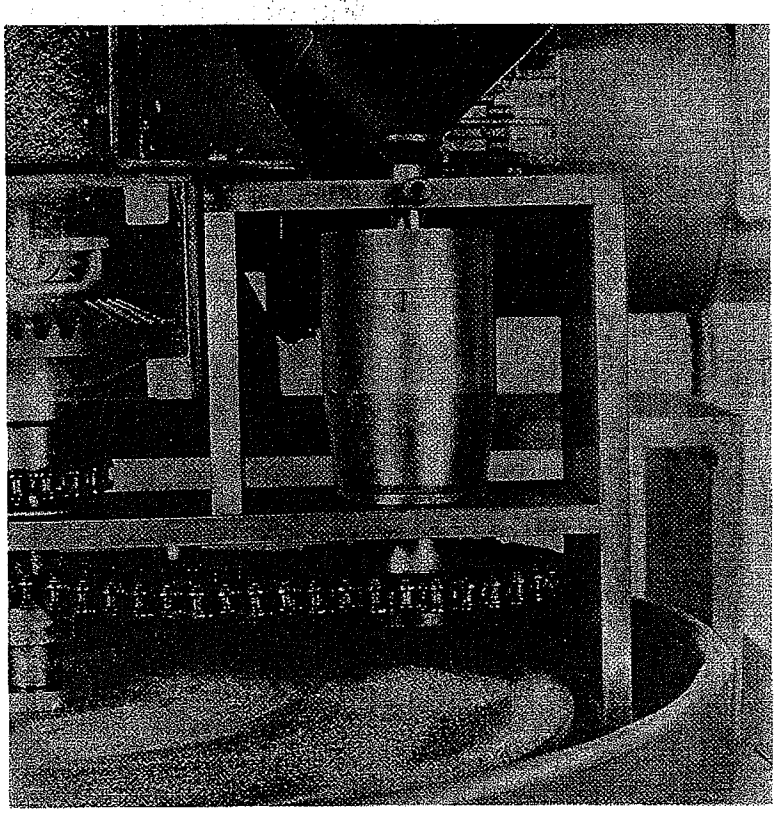


Figure 12 - Detail of distributor of abrasive grits.

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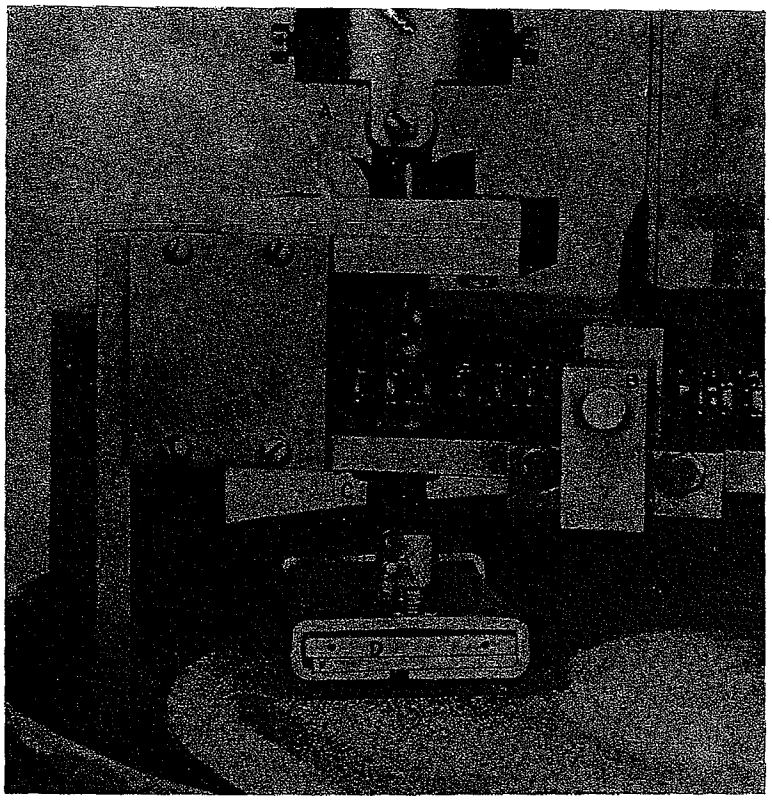


Figure 13 - Detail of cam and specimen holder.

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